

In today's competitive industrial and corporate environment, organizations constantly seek efficient methods to optimize workforce management and productivity tracking. One of the most critical aspects of this is maintaining accurate employee attendance records and monitoring work hours. Traditional methods such as manual logbooks, RFID cards, or biometric systems are still widely used but come with multiple limitations.

Manual systems are prone to human errors, manipulation, and buddy punching. Biometric devices require physical contact and regular maintenance and may not function reliably in outdoor or rugged environments. RFID systems, while touchless, require the user to scan their card actively, which may not always happen consistently. Moreover, none of these methods provide real-time data access or automatic work-hour calculation without additional software integration.

To address these drawbacks, there is a growing need for a fully automated, contactless, and accurate attendance tracking system that can work efficiently in real-time and support remote data access. The **BLE-Based Employee Tracking System** is designed to fulfill this need.

By leveraging BLE (Bluetooth Low Energy) technology, this system allows passive tracking of employee presence without requiring any manual intervention. Employees carry BLE beacons that automatically broadcast signals, which are detected by an ESP32 receiver as they enter or exit the premises. This ensures accuracy, prevents misuse, and eliminates the need for employees to remember to check in or out.

Additionally, with backend support via Firebase and a centralized web dashboard, the system provides instant access to attendance logs and total work hour calculations. This enhances transparency, helps streamline payroll processing, and allows managers to monitor workforce activity even when off-site.

Given the low power consumption of BLE devices and the scalability of the system, it is also a cost-effective solution suitable for a wide range of applications—from small offices to large construction sites and manufacturing units.

Thus, this system fulfills a real and growing need for smarter, more reliable, and hands-free employee attendance management, making it an ideal fit for modern digital workplaces.